DX-SMART

DX-LR22-900T22D
Serial port application guide

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深圳大疆无线科技有限公司

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Contents

1. Introduction ..... - 6 -
1.1. A serial port basic parameters ..... - 6 -
1.2. Module default RF basic parameters ..... - 6 -
1.3. Transport mode and the AT command mode ..... - 7 -
2. PC testing tools ..... - 7 -
2.1. Pc-side testing software ..... - 7 -
3. Serial port usage ..... - 8 -
3.1. Module test minimum system ..... - 8 -
3.2. Notes for module use ..... - 8 -
3.3. Module USES operation example ..... - 9 -
3.3.1. Module to module transparent transfer ..... - 9 -
3.3.2. Fixed transmission module and module ..... - 9 -
3.3.3. Module and module broadcast transmission ..... - 10 -
3.3.4. The mode switch ..... - 11 -
3.3.4.1. Mode 1 (Configure working mode via AT command) ..... - 11 -
3.3.4.2. Way 2 (by M0 / M1 pin configuration four work modes) ..... - 11 -
3.3.5. Instructions on the AUX module working state indicator foot ..... - 12 -
4. Related AT commands are explained in detail ..... - 12 -
4.1. Command Format Description ..... - 12 -

4.2. Response format specification ..... - 13 -
4.3. The AT command for example ..... - 13 -
4.4. List of AT commands ..... - 14 -
5. Explanation of the AT command ..... - 15 -
5.1. Basic commands ..... - 15 -
5.1.1. Enter or exit the AT command mode ..... - 15 -
5.1.2. Test the instruction ..... - 16 -
5.1.3. Software restart ..... - 16 -
5.1.4. factory data reset ..... - 16 -
5.2. Serial port parameters ..... - 16 -
5.2.1. Settings \ Query - Serial Baud rate ..... - 16 -
5.2.2. Set \ Query - serial port check bit ..... - 17 -
5.3. LORA parameters ..... - 17 -
5.3.1. Query configuration information ..... - 17 -
5.3.2. Settings \ Query - One click to configure module air rate and communication distance ..... - 18 -
5.3.3. Set \ Query - Transmission mode ..... - 20 -
5.3.4. Set \ Query - working mode ..... - 21 -
5.3.5. Setup \ Query - Hardware control pin status ..... - 22 -
5.3.6. Set \ Query - Working channel ..... - 23 -
5.3.7. Settings \ Query - Device Address ..... - 24 -

5.3.8. Settings \ Query - Module Key switch ..... - 25 -
5.3.9. Set up key - module ..... - 25 -
5.3.10. Set \ query-subpacket length ..... - 26 -
5.3.11. Settings \ Query - Packet RSSI ..... - 26 -
5.3.12. Set \ Query - Transmit power ..... - 27 -
5.3.13. Set \ query -LBT status ..... - 27 -
5.3.14. Set \ query-lbt listening threshold ..... - 28 -
5.3.15. Query - current channel noise level ..... - 28 -
5.3.16. Set \ Query - IQ flip ..... - 28 -
5.3.17. Set \ query-CRC checksum ..... - 29 -
5.4. List of error codes ..... - 29 -
6. Value-added services ..... - 30 -

Image Index

Figure 1: Computer side serial port software Figure ..... - 7 -
Figure 2: module figure ..... - 8 -

1. Introduction

DX-LR22-900T22D is a low power LoRa module, is developed by shenzhen DX-SMART co., LTD. For intelligent wireless data transmission, using LLCC68 chip, chip internal integration output maximum 22 DBM power amplifier, anti-jamming performance and further improve communication distance, Has the advantages of low power consumption, high performance and long distance.

1.1. A serial port basic parameters

- Module serial default parameters: 9600 BPS / 8 / n / 1 (baud rate/data/no calibration/stop bit)

1.2. Module default RF basic parameters

- Module rate level: LEVEL2 stores (2148 BPS)
- Module mode switching mode: Mode 1 (AT command control)
- Module subpacket length: 230 bytes
- Module packet RSSI: off
- Module key: 12345
- Module frequency band: 915.15MHz
- Module transmit power: 22dBm
- Module working mode: transparent transmission
- Module power consumption mode: high aging mode
- Module address: FFFF
- The CAD module testing time: 4 s

1.3. Transport mode and the AT command mode

- Transfer mode: module after power on, is the transmission mode, can begin to transmit data at this time.
- The AT command mode: in transmission mode, the use of "+ + +" switch to the AT command mode, can respond to the AT command. To enter the transmission mode, you need to send "+++" to exit the AT command mode.

2. PC testing tools

2.1. Pc-side testing software

Install Uart Assistant computer serial port software, open the serial port software and select the corresponding COM port, install the default parameter configuration of the serial port software, that is, 9600bps/8/n/1 (baud rate/data bit/no parity/stop bit), fill in the corresponding command, and be sure to add carriage return and line feed (you can directly press the Enter key) or check "Add carriage return and line feed", and then send the command, as shown below:

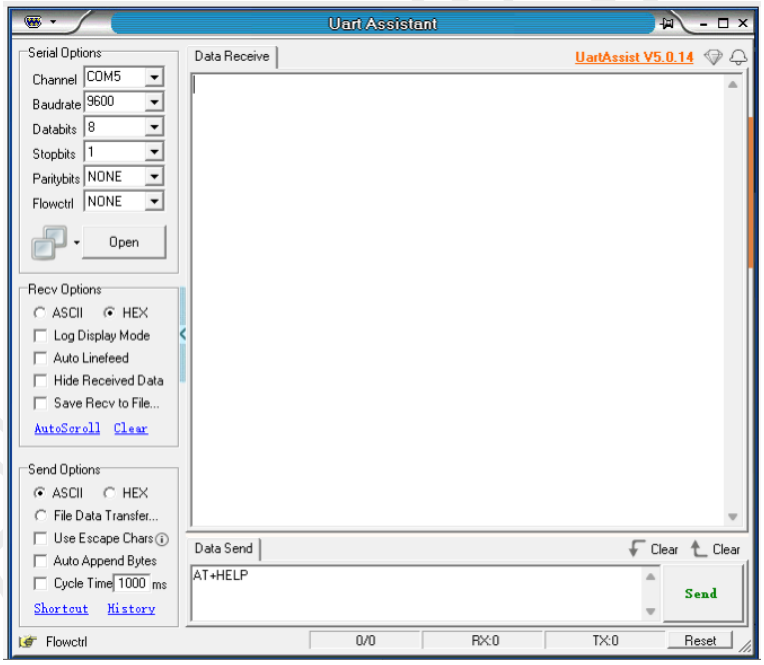


Figure 1: the PC serial port software diagram1

3. Serial port usage

3.1. Module test minimum system

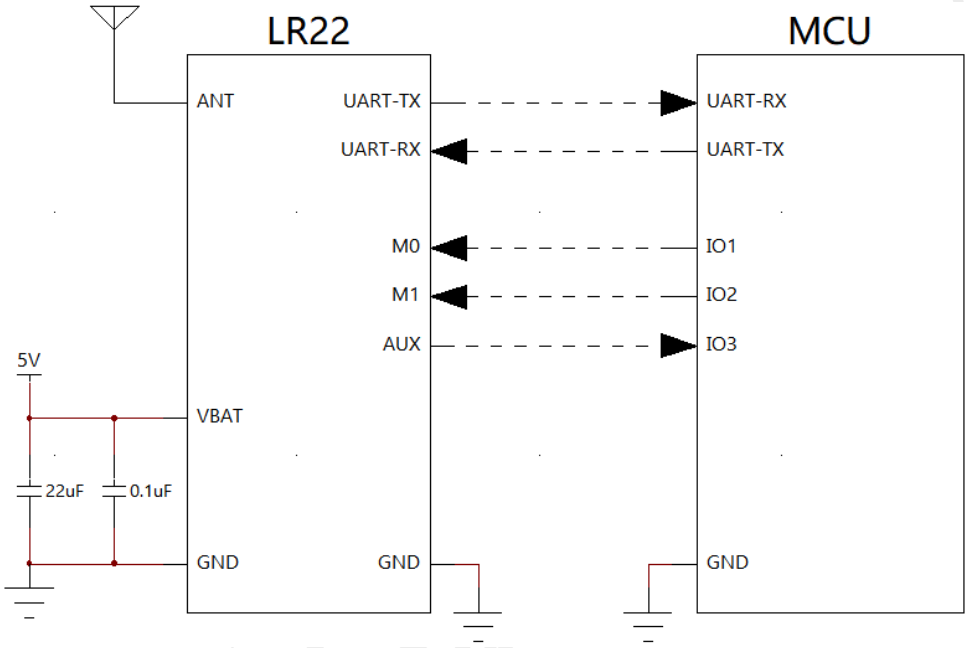


Figure 2: the module's minimum system diagram2

3.2. Notes for module use

- For LoRa modulation mode, the characteristics of information transmission delay longer, suggest customers not under low airspeed for large amount of data transmission, in order to avoid data loss caused by the accumulated data communication exception.
- When the module from the other mode is switched to sleep mode, if there are data has not been processed, the module will these data (including received and sent) after processing, can enter into sleep mode.
- When in sleep mode into the other mode, the module will start configuration parameters, the configuration process, AUX maintain a high level; After the completion of the output low level, so suggest user testing AUX level changes.

3.3. Module USES operation example

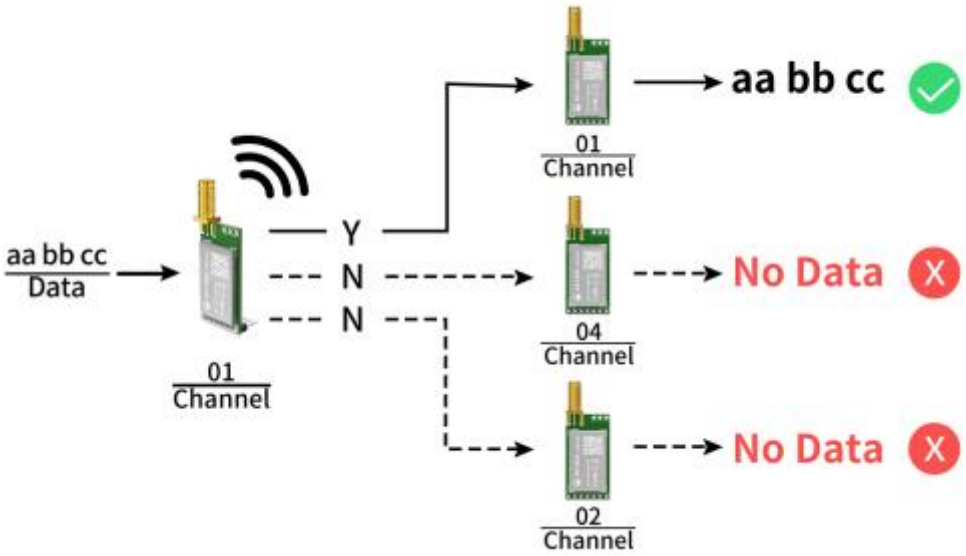
3.3.1. Module to module transparent transfer

With two DX - LR22-900 t22d module as an example, the module a and module b respectively

1. A and b connected a serial port and power supply module
2. Module a and b into the AT mode: + + +
3. Modules a and b are set to transparent transmission mode: AT+MODE0
4. Module a and b level of configuration the same rate: the AT + LEVEL2 stores
5. Module a and b: configuration of the same channel AT + CHANNEL01
6. Module a and b to restart command takes effect: the AT + RESET
7. Data transfer: One module sends data, and the other module can receive data

Note: (1) lora is half-duplex protocol, so only a module to send at a time

(2) Transparent transmission mode, need two same module rate level, channel, to transmit data to each other



3.3.2. Fixed transmission module and module

With two DX - LR22-900 t22d module as an example, the module a and module b respectively

1. A and b connected a serial port and power supply module
2. Module a and b enter AT mode: +++
3. Module a and b is set to the fixed-point transport pattern: the AT + MODE1
4. Module a and b level of configuration the same rate: the AT + LEVEL2 stores

5. Module a and b to restart command takes effect: the AT + RESET

6. Data transfer: two module choice HEX format to send, and according to the specific data format data should be transmitted between them

Note: (1) lora is half-duplex protocol, so only a module to send at a time

(2) the fixed-point transport pattern belong to private agreement, need two modules rate of the same grade, and according to the following data format, to send and receive data properly:

Data format: the address of the receiver (hexadecimal, two bytes) + the receiver channel (hexadecimal, 1 byte) + data (hexadecimal)

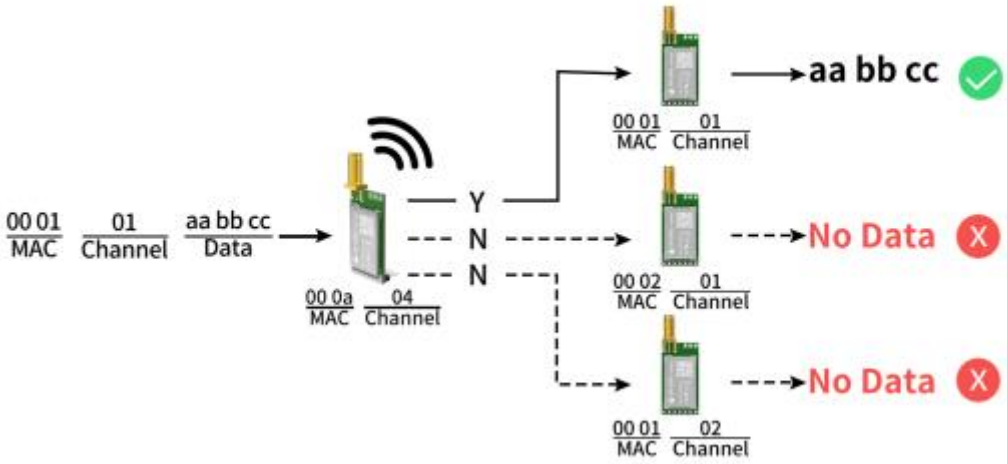
For example:

(Module a is the sender; module b is the receiver; module b has address 0001 and channel 01)

Module to send data to the module b aabbcc:

(1) Module a selects HEX format to send

(2) Module a send data content for: 00 01 01, 61, 61, 62, 62, 63, 63



3.3.3. Module and module broadcast transmission

Take two DX-LR22-900T22D modules as an example, which are module a and Module b

1. Module a and b are connected to the serial port and power supply

2. Module a and b into the AT mode: + + +

3. Module a and b is set to broadcast transmission mode: the AT + MODE2

4. Modules a and b are configured with the same rate level: AT+LEVEL2

5. Module a and b to restart command takes effect: the AT + RESET

6. Data transmission: The two modules choose HEX format to send, and transfer data to each other in accordance with the specific data format

Note: (1) lora is half-duplex protocol, so only a module to send at a time

(2) radio transmission mode belongs to private agreement, need two modules rate of the same grade, and according to the following data format, to send and receive data properly:

Data format: the receiver channel (hexadecimal, 1 byte) + data (hexadecimal)

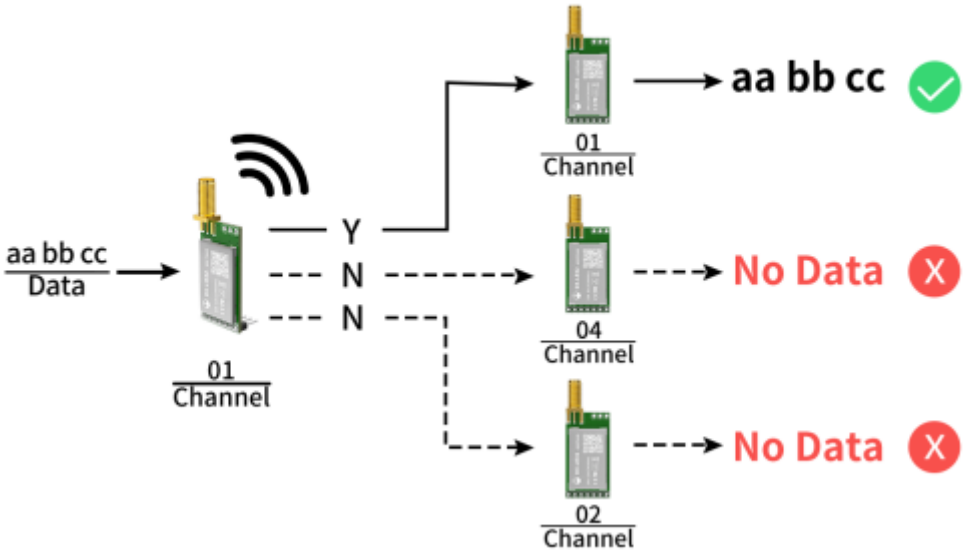
For example:

(Module a is the transmitter; module b is the receiver, and module b has 01 channel)

Module to send data to the module b aabbcc:

(1) Module a selects HEX format to send

(2) Module a sends the data content as: 01 61 61 62 62 63 63



3.3.4. The mode switch

3.3.4.1. Mode 1 (Configure working mode via AT command)

+ + + into the AT command mode, the configuration is required with the following command mode:

- High Aging mode: AT+SLEEP2
- Air Wake Mode: AT+SLEEP1
- Sleep mode: the AT + SLEEP0

3.3.4.2. Way 2 (by M0 / M1 pin configuration four work modes)

When the module command AT + SWITCH = 1, by M0 / M1 pin SWITCH module working mode, working pattern corresponding to the table below:

M0 input level	M1 input level	Operating mode

High level	High level	Sleep mode
High level	Low level	Air wake model
Low level	High level	AT mode
Low level	Low level	High aging model

Notes

1、Command AT + SWITCH details please refer to 5.3.5 Settings \ query - hardware control pin.
2、The command AT + SWITCH = 1, M0 / M1 pins for internal weak pull-up, by M0 / M1 pin SWITCH module working mode, the impending, please be sure to avoid pin to prevent abnormal mode switches.

3.3.5. Instructions on the AUX module working state indicator foot

Pin name	Pin number	Description	Pin output level	Instructions
AUX	5	Indication of module working status	High level	Data transmitting/data receiving/working mode switching
			Low level	Data send complete/data receive complete/work mode switch complete

Remarks

When the module detects the received data, AUX will output a high level 2-3ms in advance, which is used to prompt the external MCU to be ready to receive the data.

4. Related AT commands are explained in detail

4.1. Command Format Description

AT+Command<param1, param2, param3> <CR><LF>

- All commands begin with AT and end with <CR><LF>. In the table that shows commands and responses in this document, <CR><LF> is omitted and only commands and responses are displayed.
- All AT command characters are uppercase.
- < > for optional content, if there are multiple parameters in command, separated with a comma ", ", actually does not include the brackets in the command.
- <CR> is the carriage return character \r, which is 0X0D in hexadecimal.
- <LF> is the newline character \n, which is 0X0A in hexadecimal.
- Instruction execution is successful, returns the corresponding command ends with a OK, failure return EEROR = < >, "< >" content for the corresponding error code (see 5.4).

4.2. Response format specification

+ Indication < = param1, param2, param3 > < CR > < LF >

- Response instructions with a plus sign "+", < CR > < LF > end
- Is equal to "=" in response to the parameters
- If there are multiple parameters in response to the parameters, you will be separated with a comma ","

4.3. The AT command for example

Example: modify LoRa equipment baud rate to 9600

Send: AT + BAUD3

Returns: OK

4.4. List of AT commands

instruction	Features	Instructions
+++	Enter or exit the AT command mode	Power on defaults to transmission mode
AT	Test instructions	Used to test the serial port
AT+RESET	Software restart	-
AT+DEFAULT	factory data reset	-
AT+BAUD	Set \ Query serial port baud rate	Default: 3 (9600)
AT+PARI	Set \ query check digit serial port	Default: 0 (no validation)
AT+HELP	Query configuration information	-
AT+LEVEL	Set \ Query module air rate and communication distance	Default: 2
AT+MODE	Set \ Query transmission mode	Default: 0 (transparent)
AT+SLEEP	Set \ Query work mode	Default: 2 (high aging mode)
AT+SWITCH	Set \ Query hardware control pin status	Default: 0 (off)
AT+CHANNEL	Set \ Query working channel	Default: 41
AT+MAC	Set \ Query device address	Default: ff,ff
AT+OPENKEY	Set \ Query module key switch	Default: 1 (on)
AT+KEY	Set the key module	Default: 12345
AT+PACKET	Settings \ query length of the subcontract	Default: 3 (230 bytes)
AT+DRSSI	Set \ Query packet RSSI	Default: 0 (off)
AT+POWE	Set \ Query transmit power	Default: 22
AT+LBT	Set \ Query LBT status	Default: 0 (off)

AT+LRSSI	Set \ query LBT listening threshold	Default: -100
AT+ERSSI	Query the current channel noise level	-
AT+IQ	Set \ Query IQ flip	Default: 1 (on)
AT+CRC	Set \ query CRC validation	Default: 1 (open)

5. Explanation of the AT command

5.1. Basic commands

5.1.1. Enter or exit the AT command mode

Features	Instructions	Response	Instructions
Enter or exit the AT command mode	+++	Entry AT or Exit AT Power On	Entry AT: enter the AT command mode Exit AT: Exit the AT command mode Power-on is in transmission mode by default

Notes:

1、It will be reset automatically when you exit the AT command mode.
2、The command is not saved when power fails.

5.1.2. Test the instruction

Function	Instructions	Response	Instructions
Testing	AT	OK	

5.1.3. Software restart

Features	Instructions	Response	Instructions
Software restart	AT+RESET	OK Power On	

5.1.4. factory data reset

Features	Instructions	Response	Instructions
factory data reset	AT+DEFAULT	OK Power On	

5.2. Serial port parameters

5.2.1. Settings \ Query - Serial Baud rate

Features	Instructions	Response	Notes
Query the baud rate	AT+BAUD	+BAUD=<baud>	<baud> The baud rate corresponds to the serial number 1:2400 5:38,400 2:4800 6:57,600 3:9,600 7:115,200 4:19200 Default: 3(9600)
Set baud rate	AT+BAUD<baud>	+BAUD=<baud> OK	

Remarks:

After setting this instruction, it should be restarted to take effect.

5.2.2. Set \ Query - serial port check bit

Features	Instructions	Response	Instructions
Query check digit serial port	AT+PARI	+PARI=<param>	< param> sequence number 0: No validation 1: odd check 2: Even check Default: 0
Set the serial port check bit	AT+PARI<param>	+PARI=<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

5.3. LORA parameters

5.3.1. Query configuration information

Features	Instructions	Response	Instructions

Query basic module configuration information	AT+HELP	========== LoRa Parameter: +VERSION=<version> MODE:<mode> LEVEL:<level> SLEEP:<sleep> Frequency:<frequency> MAC:<mac> CRC:<crc> IQ:<iq> Power:<power> ==========	LoRa Parameter: LoRa parameter <version>: Version <mode> : Transmission mode <level> : Air rate configuration "Sleep > : working mode <frequency> : frequency of work MAC > < : device address <crc> : CRC check <iq> : IQ flips <power> : Transmit power

Examples:

Query module basic information

Send: AT+HELP

Return to: ==========

LoRa Parameter:

+ VERSION = V1.2.3

MODE:0

LEVEL:2 >> 2149bps

SLEEP:2

Frequency:915150000hz >> 41

MAC:ff,ff

CRC:1(true)

IQ:1(true)

Power:22dBm

==========

5.3.2. Settings \ Query - One click to configure module air rate and communication distance

Features	Instructions	Response	Instructions
The query module parameters	AT+LEVEL	+LEVEL =<param>	< param > : 0 to 7, Air speed and communication range configuration, with eight gears Default value: 2
Set the module parameters	AT+LEVEL<param>	+LEVEL =<param> OK	

Notes:

1, can choose according to their own communication distance and quantity of the data of different gear (data volume and distance can refer to the table below). Air rate, the greater the characters can be sent the amount of data the faster, the shorter the distance communication. The air rate is inversely proportional to the distance characters.

2, the command will RF bandwidth, RF coding rate, spread spectrum factor has been set, can be used directly.

3. The LEVEL of transmitting equipment and receiving equipment should be consistent to receive and send data.

After 4, set this directive to restart to take effect.

Note: the following table for the configuration parameters under different gear, the following open visual range, for reference only, actual distance will be subject to the measured.

LEVEL(gear)	Spread spectrum factor (SF)	BW(bandwidth KHz)	CR (coding rate)	Air symbol rate (bit/s)	Open visible distance (Km)
0	11	125	4/8	336	8.0
1	11	250	4/5	1075	-
2	11	500	4/5	2148	-
3	8	250	4/5	6250	-
4	8	500	4/6	10417	-
5	7	500	4/6	18229	-
6	6	500	4/5	37500	-
7	5	500	4/5	62500	-

5.3.3. Set \ Query - Transmission mode

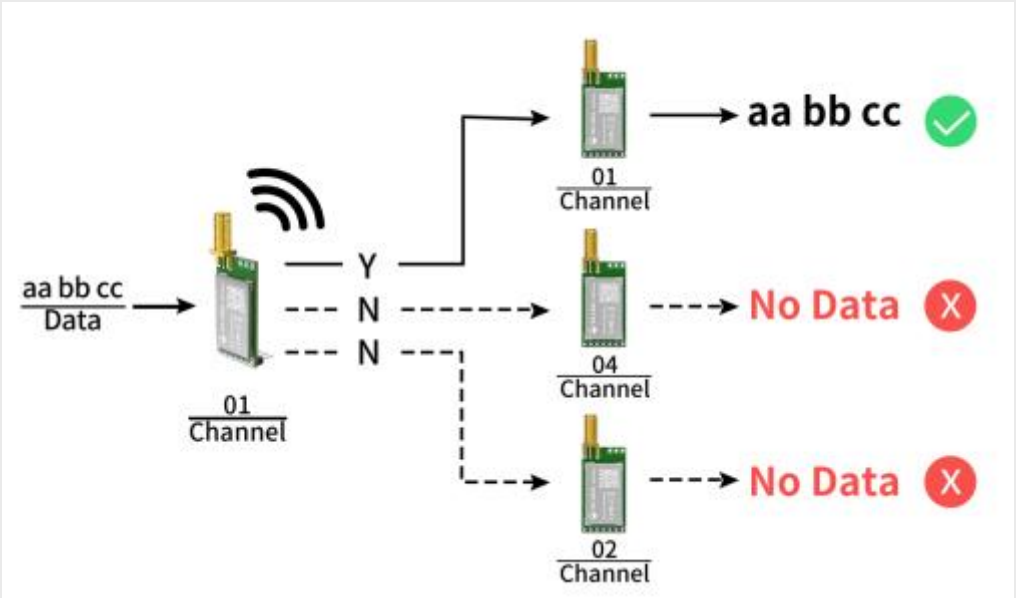
Features	Instructions	Response	Instructions
Query transmission mode	AT+MODE	+MODE=<param>	param: 0,1,2 0: transparent transmission 1: Fixed-point transmission 2: Broadcast transmission Default setting: 0
Set transmission mode	AT+MODE<param>	+MODE=<param> OK	

Notes:

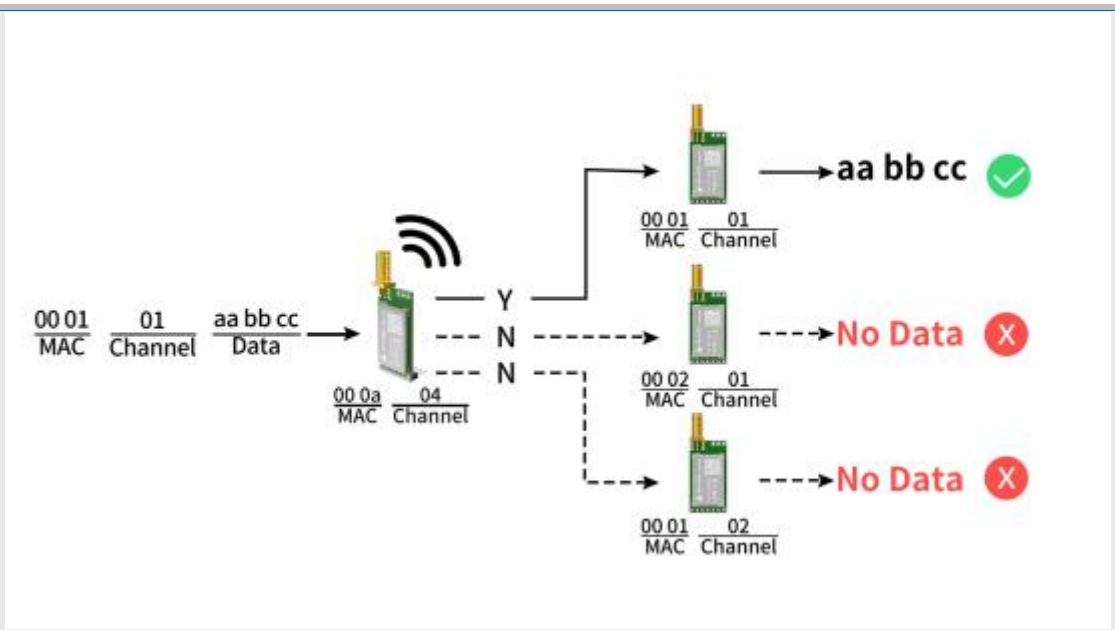
1、After setting this instruction, it should be restarted to take effect.
2、Transparent data transmission formats: send data directly
3、Fixed-point transmission data format: device address (hexadecimal, two bytes) + channel number (hexadecimal, one byte) + data (hexadecimal)
4. Broadcast transmission data format: channel number (hexadecimal, one byte) + data (hexadecimal)

Examples:

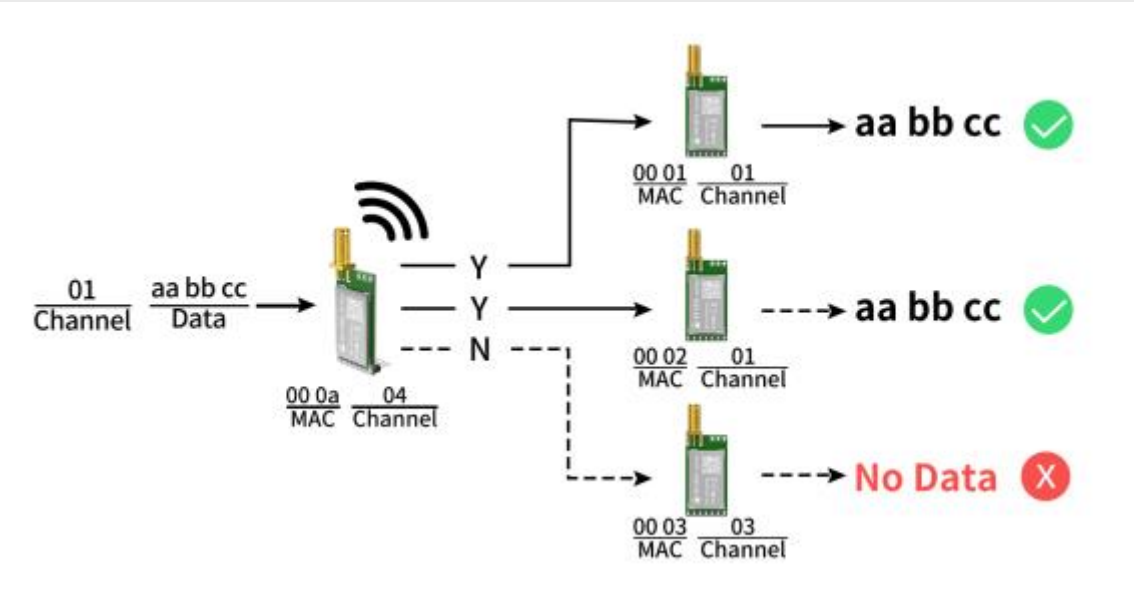
1、Transparent transmission: send data directly



2、Fixed transmission: the address of the receiving module is 0001, channel 01; Emission module to send data to aabbcc, then send the data content is: 000101 aabbcc (hex: 00 01 01, 61, 61, 62, 62, 63, 63)



3、Broadcast transmission: receiving module for channel 01, emission module to send data to aabbcc, sending data content as: 01 aabbcc (hex: 01, 61, 61, 62, 62, 63, 63)



5.3.4. Set \ Query - working mode

function	Instructions	Response	Instructions
Query work mode	AT+SLEEP	+SLEEP=<param>	< param > serial number 0: Sleep mode 1: Air wake-up mode 2: high aging model
Set the work mode	AT+SLEEP<param>	+SLEEP=<param> OK	

Default value: 2

Notes:

1、Sleep mode: this mode, MCU and radio frequency to enter a dormant state.

Method one: use a serial port to sleep and wake up, send the AT + SLEEP0 into sleep mode, wake up when sending to awaken AT + WAKEUP or any data, wake up after the module is in a state of the AT command.

Method 2: use, M0, M1 feet to sleep and wake up, send the AT + SWITCH1 open M0, M1 mode switch, module automatically restart, = 1 set M0, M1 = 1, enter the sleep mode, M0, M1, of arbitrary switching module out of sleep mode.

2、Air wake mode:

A, the mode, the module for four seconds A cycle test CAD (overall sleep time is: 4 s minus CAD test time), such as module detected data, will enter the receive mode, after receiving the data, automatically entered into dormancy. During the sleep period, the RF will sleep, and the MCU will not sleep.

B, using air wake model, the receiver and the sender should wake up in the air mode, can send and receive data.

C, the model can be written to save.

Note: CAD explanation: LoRa CAD (Channel Activity Detection) is a technology used to detect channel activity in LoRa network. It is used to determine whether there is any activity on the specified physical channel (such as other equipment), to help equipment to choose the appropriate time to sending and avoid collisions.

3、High aging mode: the mode, the module has been at the receiving state, ready to receive the data of other equipment. When the module of serial port to receive the master data, namely state, switch to launch the data launch out, after the completion of the launch, switch back to the receiving state.

4、Settings take effect after the directive to restart.

5.3.5. Setup \ Query - Hardware control pin status

Features	Instructions	Response	Instructions
Query the hardware control pin status	AT+SWITCH	+SWITCH=<param>	<param> : 0,1 0: close 1: Open Default: 0 (off)
Sets the hardware	AT+SWITCH<param>	+SWITCH=<param>	

control pin state	OK

Remarks:

1、After setting this instruction, it should be restarted to take effect.

2、When SWITCH=1, the working mode can be selected through the module hardware pins M0, M1: (below 0 is low level, 1 is high level)

M0 = 0, M1 equals zero, the module in high aging model;

M0 = 1, when the M1 = 0, module wake up in the air mode;

When M0=0, M1=1, the module is in AT mode;

When M0=1, M1=1, the module is in sleep mode.

3、When the module enters the sleep mode through the pin control, it cannot wake up through the serial port or the command, and must exit the sleep mode through M0 and M1.

4、When the SWITCH = 0, can pass the AT + SLEEP this instruction to select the work mode.

5.3.6. Set \ Query - Working channel

Functions	Instructions	Response	Instructions
Query the working channel	AT+CHANNEL	+ CHANNEL=<param>	Param: 00-63 (hexadecimal) With the opening of the 850.15 Mhz, The channel interval is 1Mhz The default Settings: 41
Set the working channel	AT+CHANNEL <param>	+CHANNEL=<param> OK	

Notes:

1, this module is set with 100 general channels, if you need more, please contact our company.

2. After the instruction is set, it needs to be restarted.

3. When multiple receiving devices are too close to the transmitting device, it may lead to the receiving device of different channels can receive data, so the distance between the transmitting device and the receiving device is required to be as far as possible.

Note: the following table for different channel of the frequency control, unit: Mhz.

Channel	Operating frequency	channel	Operating frequency	Channel	Operating frequency	Channel	Operating frequency

band		band		band		band	
00	850.15	19	875.15	32	900.15	4B	925.15
01	851.15	1A	876.15	33	901.15	4C	926.15
02	852.15	1B	877.15	34	902.15	4D	927.15
03	853.15	1C	878.15	35	903.15	4E	928.15
04	854.15	1D	879.15	36	904.15	4F	929.15
05	855.15	1E	880.15	37	905.15	50	930.15
06	856.15	1F	881.15	38	906.15	51	931.15
07	857.15	20	882.15	39	907.15	52	932.15
08	858.15	21	883.15	3A	908.15	53	933.15
09	859.15	22	884.15	3B	909.15	54	934.15
0A	860.15	23	885.15	3C	910.15	55	935.15
0B	861.15	24	886.15	3D	911.15	56	936.15
0C	862.15	25	887.15	3E	912.15	57	937.15
0D	863.15	26	888.15	3F	913.15	58	938.15
0E	864.15	27	889.15	40	914.15	59	939.15
0F	865.15	28	890.15	41	915.15	5A	940.15
10	866.15	29	891.15	42	916.15	5B	941.15
11	867.15	2A	892.15	43	917.15	5C	942.15
12	868.15	2B	893.15	44	918.15	5D	943.15
13	869.15	2C	894.15	45	919.15	5E	944.15
14	870.15	2D	895.15	46	920.15	5F	945.15
15	871.15	2E	896.15	47	921.15	60	946.15
16	872.15	2F	897.15	48	922.15	61	947.15
17	873.15	30	898.15	49	923.15	62	948.15
18	874.15	31	899.15	4A	924.15	63	949.15

5.3.7. Settings \ Query - Device Address

Features	Instructions	Response	Instructions

Looking up device addresses	AT+MAC	+MAC=<param>,<param>	param: Hexadecimal, one byte Default Settings: ff,ff
Set device address	AT+MAC<param>,<param>	+MAC=<param>,<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

Set the module address to 0a01
Send: AT+MAC0a,01
Return: +MAC=0a,01
OK

5.3.8. Settings \ Query - Module Key switch

Features	Instructions	Response	Instructions
Query the module key switch	AT+OPENKEY	+OPENKEY=<param>	<param> 0,1 0: Close key 1: Turn key on Default value: 1
Set the module key switch	AT+OPENKEY<param>	+OPENKEY=<param> OK	

Notes:

1、After the instruction is set, it needs to be restarted to take effect.
2、The AT + OPENKEY = 0, can turn off the key function, but are not clear set of keys.
3、The OPENKEY of both sender and receiver must be consistent, otherwise the sender and receiver will be abnormal.

5.3.9. Set up key - module

Features	Instructions	Response	Instructions
Set the key module	AT+KEY<param>	+KEY=<param> OK	<param>0~65535 Default value: 12345

Remark:

After 1, set this directive to restart to take effect.
2, the instruction can only be set not query.
3, and the keys must be consistent, otherwise it will send and receive abnormal.
4. If you need to restore the key value to the DEFAULT, you need to send AT+DEFAULT to restore all parameters of the module to the default value.

5.3.10. Set \ query-subpacket length

function	Instructions	Response	Instructions
Query the subcontract length	AT+PACKET	+ PACKET=<param>	Param: 0 ~ 3 0:32 bytes 1:64 bytes "8 bytes And zero bytes Default: 3 (230 bytes)
Set the subpacket length	AT+PACKET<param>	+PACKET=<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

5.3.11. Settings \ Query - Packet RSSI

Features	Instructions	Response	Instructions
Query packet RSSI	AT+DRSSI	+DRSSI=<param>	<param> : 0,1 0: close 1: Open Default: 0 (off)
Set packet RSSI	AT+DRSSI<param>	+DRSSI=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.

2、Add the received signal strength to the received packet.

Examples:

Sender: 31 32 33 34 35
Receiver: 31 32 33 34 35 AB
Where the last bit "AB" is the RSSI value detected in real time, which will change with the strength of the signal.
The RSSI value is calculated as follows (using the value AB as an example) :
Actual signal strength =- (0xFF-0xAB) =- (255-171) =-84 dBm
Therefore, the received signal strength for this time is -84 dBm. Note that the RSSI value may be different for each communication, and it depends on the current wireless signal quality. 0xFF in the above formula is a fixed value.

5.3.12. Set \ Query - Transmit power

Features	Instructions	Response	Instructions
Query transmit power	AT+POWE	+POWE=<param>	param: 0-22dBm (take integer values) Default setting: 22dBm
Set transmit power	AT+POWE<param>	+POWE=<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

Examples:

Modify the transmit power to 10dBm
Send: AT+POWE10
Return: +POWE=10
OK

5.3.13. Set \ query -LBT status

Features	Instructions	Response	Instructions

Check LBT status	AT+LBT	+LBT=<param>	<param> : Sequence number 0: closed 1: Open Default: 0
Set the LBT status	AT+LBT<param>	+LBT=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.
2、When the LBT is set to 1, the channel noise will be monitored first. When the noise is greater than the threshold, the channel noise will be monitored continuously, and it will be sent directly after 2S.

5.3.14. Set \ query-lbt listening threshold

Features	Instructions	Response	Instructions
Query the LBT listening threshold	AT+LRSSI	+LRSSI=<param>	<param> : -255 to 0 The current LBT listening threshold Default value: -100
Set the LBT listening threshold	AT+LRSSI<param>	+LRSSI=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.
2、Changing the parameter will change the LBT listening threshold.
3. This default value is obtained from our company's test in a specific environment, and the user needs to set the noise level when the channel is idle through AT+ERSSI.

5.3.15. Query - current channel noise level

Features	Instructions	Response	Instructions
Query the current channel noise level	AT+ERSSI	+ERSSI= <param>	<param> : Current channel noise level

Remarks:

Enquiries only.

5.3.16. Set \ Query - IQ flip

Features	Instructions	Response	Instructions
Query IQ flip status	AT+IQ	+IQ=<param>	<param> : 0 (off), 1 (on) Default: 1
Set the IQ flip state	AT+IQ<param>	+IQ=<param> OK	

Notes:

1、After setting this instruction, it should be restarted to take effect.
2、The sender and receiver must ensure the same IQ value to communicate normally.

5.3.17. Set \ query-CRC checksum

Features	Instructions	Response	Instructions
Query the CRC check status	AT+CRC	+CRC=<param>	<param> : 0 (off), 1 (on) Default: 1
Set the CRC check status	AT+CRC<param>	+CRC=<param> OK	

Notes:

After setting this instruction, it should be restarted to take effect.

5.4. List of error codes

The details of the error code in EEROR=<> are as follows:

Return value	Error Message Description
104	Invalid instructions
105	Invalid parameters
106	Other errors

6. Value-added services

In order to meet the various functional requirements of customers, our company can provide the following technical value-added services:

- Module program customization, such as: IO function port customization, AT instruction customization, etc.
- Module PCB hardware customization, can be customized to the hardware requirements of customer needs.
- A variety of program customization, can be customized according to customer needs, a full set of software and hardware solutions.
- A full set of networking solutions customization, can be customized according to customer needs, a full set of networking, gateway solutions.

If you have the above customized requirements, please contact our business personnel directly.

深圳大夏龙雀科技有限公司